REMOTE SENSING IMAGERY SCENE CLASSIFICATION BASED ON SPIKING NEURAL NETWORK

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ABSTRACT

In order to overcome the large computational cost of deep neural networks (DNNs), spiking neural networks (SNNs) have been proposed, which are more biologically reasonable. It has the potential to achieve energy efficiency while maintaining performance comparable to DNNs. Although SNNs have achieved good results on the MNIST and CIFAR10 data sets, its potential in remote sensing has not been studied and explored. This paper adopts the idea of converting a trained DNN into an SNN, and proposes a multi-bit-based SNN, and introduces channel normalization (channel-norm) to replace the previous layer normalization to achieve remote sensing imagery scene classification tasks. By achieving multi-bit spiking and channel-norm in the way of DNN conversion to SNN, the SNN proposed in this paper achieves lossless conversion on the UC Merced data set and WHU-RS data set.

Index Terms— DNNs, SNNs, multi-bit spiking, channel-norm, remote sensing imagery scene classification

1. INTRODUCTION

Remote sensing imagery classification tasks are the basis for achieving remote sensing imagery semantic understanding, imagery target segmentation, etc. Currently, deep neural networks (DNNs) represented by convolutional neural networks (CNNs) have achieved remarkable results in the field of remote sensing imagery scene classification[1][2][3][4]. As CNNs become more and more complex, they need more powerful computing platforms to implement, which limits their practical applications.

In recent years, due to its event-driven nature and low energy consumption, spiking neural networks (SNNs) have been proposed as the third generation of neural networks[5]. SNNs are fundamentally different from traditional neural networks. SNNs transmit information through the precise timing of spike trains composed of a series of discrete spiking values, rather than a series of continuous values. A recent demonstration by TrueNorth[6] showed that CNNs with more than 1 million neurons can be implemented on a set of chips that consume only a few hundred mW. If the spiking form of DNNs can be implemented on these high-efficiency platforms while still producing good accuracy, this will be crucial for achieving fast and efficient real-time remote sensing imagery scene classification tasks.

Currently, how to train SNNs are still challenging due to the complex dynamics and non-differentiable operations of spiking neurons. The method of converting DNNs into SNNs as an alternative method has been extensively studied in recent years [7][8][9]. The method narrows performance gap between SNN and DNN by training a traditional neural network and building a conversion algorithm to map the weights to an equivalent SNN. As far as we know, the potential of SNNs in scene classification has not been explored. Therefore, we design a scene classification network based on SNN to take full advantage of its potential.

In this paper, we propose an SNN model for scene classification. Since the previous SNN performs poorly on this task, we think there are two reasons. The first is insufficient spiking rates activation, and the second is excessive propagation errors. The previous architecture proposed by Diehl et al.[7] is designed for simple MNIST and CIFAR10 data sets, and the number of network layers is relatively shallow, so the above-mentioned error is not obvious layer by layer. In real task, remote sensing images contain more complex textures and information, and a deep conversion network needs to be built to achieve scene classification. With this in mind, we design a multi-bit spiking model and replace the previous layer normalization with channel normalization. Experimental results show that it can perform well on complex texture remote sensing data sets.

The remainder of this paper is organized as follows. Section 2 describes architectural overview of our model. Section 3 provides experimental results and discussions, and Section 4 concludes the paper. References are displayed at the end.

2. METHOD

The overall architecture of our model consists of two parts. The architecture of our DNN is shown in Fig. 1 and the architecture of our SNN is shown in Fig. 2.

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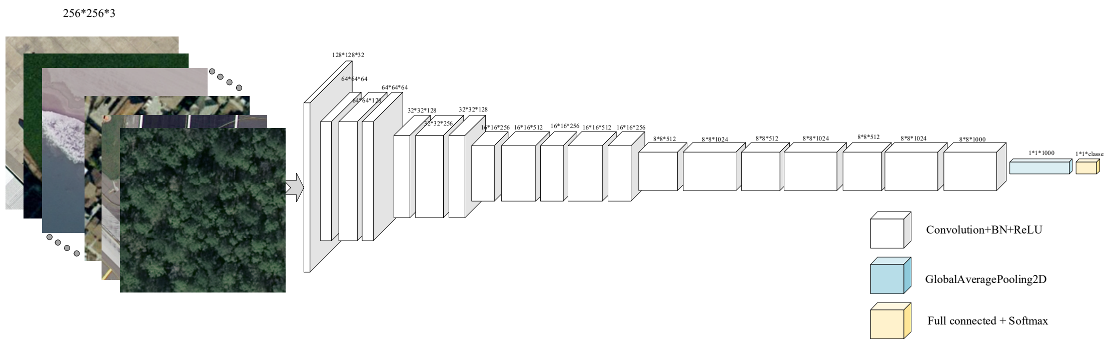

Fig. 1. The architecture of our deep neural network.

2.1. Deep neural network architecture

We propose a new network architecture to train our DNN weights, including 19 convolutional layers, 1 global average pooling layer and 1 fully connected classification layer. The convolutional layers use 3×3 or 1×1 filters, and 1×1 filters are used to compress feature representation between 3×3 convolutions. We use convolutional layers with stride 2 to replace convolutional layers with stride 1 and max-pooling layers to reduce the amount of model calculations. Batch normalization (BN) method is used to improve convergence speed, making model more stable and more standardized. The architecture of our DNN is shown in Fig. 1.

2.2. Conversion of DNNs into SNNs

The network conversion composed of two different types of neurons is based on the assumption that activation values of DNN approximate firing rates of spiking neurons. We replace artificial neurons with spiking neurons and establish synaptic connections between neurons to map DNN to SNN. The spiking neuron integrates the input current until the membrane potential crosses the firing threshold, a spike is generated and the membrane potential is reset.

In Fig. 2, we illustrate the architectural overview of our conversion methods. ReLU activation function is replaced by integrate-and-fire (IF) neuron. Convolutional layer is replaced by a synaptic layer that operates similarly to convolution. Pooling layer and fully connected layer are replaced by similar synapses of these operations, and a layer of IF neuron is added to integrate spikes of these synaptic connections. For the BN layer, directly apply batch normalization method to integrate the parameters of BN layer into the parameters of convolutional layer. Weights and biases of the convolutional layer are scaled to:

$$\widehat{W} = \frac{\gamma W}{\sigma}\ ,\widehat{b} = \frac{\gamma(b-\mu)}{\sigma} + \beta \tag{1}$$

where σ is the variance, μ is the mean, γ and β are two learnable parameters.

Before we map weights in DNN to SNN, since we use

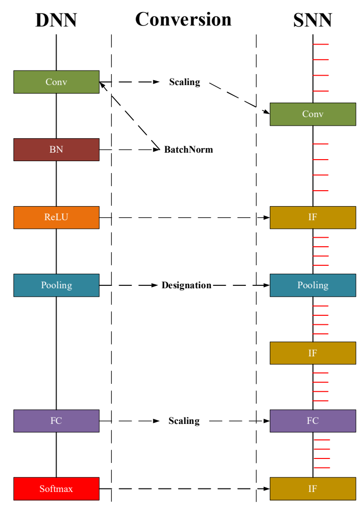

Fig. 2. Architectural overview of our conversion methods.

the ReLU function in training, the activation values of DNN can be any positive real number. Meanwhile, the range of firing rate of spiking neuron is $[0\ ,r_{max}]$, where r_{max} is the maximum firing rate of a biological neuron firing at most once in a single time step. In order to have activations in DNN matching firing rates in SNN (both in the range of [0, 1]), weights and biases in DNN are jointly normalized (layer normalization):

$$\overline{W}^{l} = W^{l}\frac{\lambda^{l-1}}{\lambda^{l}}\ ,\overline{b}^{l} = \frac{b^{l}}{\lambda^{l}} \tag{2}$$

where λ^{l} is the max 99.0% activation at layer l instead of the actual max activation, due to be susceptible to outliers.

2.3. Channel normalization

We introduce a more fine-grained normalization method, channel normalization (channel-norm)[8] instead of layer normalization (layer-norm), to achieve the fast and efficient information transmission in SNNs. This method uses the channel approach instead of the traditional hierarchical approach, and normalizes the weight by 99.0% to 99.9% of the maximum activation values. The channel-norm can be expressed as:

$$\tilde{W}_{i,j}^{l} = W_{i,j}^{l} \frac{\lambda_i^{l-1}}{\lambda_j^{l}} \ , \tilde{b}_j^{l} = \frac{b_j^{l}}{\lambda_j^{l}} \tag{3}$$

where i and j are indices of channels. Normalize the weights W in a layer l by maximum activation λ_j^l in each channel.

2.4. Multi-bit spiking

Most previous DNN-to-SNN conversion methods output spiking values is 0 or 1, which we call single-bit spiking[7]. SNN converts a series of continuous values into discrete spiking values 0 or 1. Therefore, there is the propagation error. The previous single-bit spiking is designed for simple MNIST and CIFAR10, and the number of layers of the conversion network is relatively shallow, and the propagation error is not obvious. However, remote sensing imageries contain more complex textures and information, and the deeper network is needed to achieve scene classification. We find that the accumulation of propagation errors is the key to hindering lossless conversion of deep networks. Therefore, we propose an effective method to compensate for propagation errors. We propose multi-bit spiking, which reduces the propagation error transmitted layer by layer by adding a small amount of spiking values. The method can be expressed as:

$$fire\left(V_{men}^{l}\right) = \begin{cases} N \ , if \ \ V_{men}^{l} \geq N \cdot V_{th} \\ 0 \ \ , \quad otherwise \end{cases} \tag{4}$$

$$N = int\left(\frac{V_{men}^{l} + z^{l}}{V_{th}}\right) \tag{5}$$

Where SNN neuron has a membrane potential V_{men}^{l}, which integrates its input current z^l at every time step. V_{th} is the firing threshold. In the follow-up experiment, the maximum value of N is 3, and spiking values output are 0, 1, 2, 3.

3. EXPERIMENTS AND RESULTS

3.1. Data sets and experiment settings

The original imageries of the UC Merced (UCM) data set are downloaded from the National Imagery of the United States Geological Survey [10], and then they are divided into small-scale imageries with sizes of 256×256 pixels. It contains 21 scene classes, each with 100 imageries, for a total of 2100 imageries.

The WHU-RS data set is extracted from Google Earth[11]. The data set contains 19 scene classes, about 1000 imageries, and the size of each imagery is 600×600.

For DNN, we use transfer learning method to pre-train our DNN model on ImageNet dataset, and use pre-trained network to train remote sensing datasets. we adopt data augmentation on the original data, including horizontal flip, vertical flip and random crop etc. We randomly divide the training set and the test set at a ratio of 8:2, and experimenttal results are the average results of repeated experiments.

For SNN, we use a firing threshold is 1V, an input rate is 1000Hz, the actual maximum activation value is 99.0% of the maximum activation value, and the spiking reset method uses reset by subtraction.

3.2. Experimental results and discussions

We evaluate the performance of our DNN model on UCM and WHU-RS datasets. Table 1 and Table 2 are comparisons between our DNN and other experimental results. It can be seen that on the UCM data set and WHU-RS data set, compared with other models, our proposed DNN has higher overall accuracy. The parameters of our proposed model are only 20.9M, while the parameters of the fine-tuned VGG-16 model [1] are 138M.

Table 1. Results of different methods (UCM Dataset).

Models	Accuracy (%)
CaffeNet[1]	95.02 ± 0.81
VGG-VD-16[1]	95.21 ± 1.20
MJDCNN-20[2]	95.79 ± 0.91
IFK-VGG-M[3]	96.90
Fusion by addition[4]	97.42 ± 1.79
Our DNN model	**98.81**

Table 2. Results of different methods (WHU-RS Dataset).

Models	Accuracy (%)
VGG-VD-16[1]	96.05 ± 0.91
CaffeNet[1]	96.24 ± 0.56
VLAD-VGG-VD16[3]	98.64
DCA by addition[4]	98.70 ± 0.22
Our DNN model	**99.50**

The smaller the conversion loss and the shorter the time step, the greater the correlation between SNN and DNN. Since the DNN model has been trained before the conversion, the accuracy of the DNN model will remain unchanged during the simulation process. We evaluate the performance of our SNN model on the UCM data set and WHU-RS data set. Fig. 3 and Fig. 4 are the comparison of the experimental results of our proposed SNN and previous SNN (the blue curve) methods. It can be seen that on the UCM and WHU-RS data sets, compared with the previous SNN (the blue curve), multi-bit spiking and channel-norm methods can speed up the convergence speed of the conversion method, and improve the accuracy of the conversion. Both the multi-bit spiking and channel-norm methods have the best effect when applied together, and can achieve DNN-to-SNN lossless conversion in less time steps.

Channel-norm can better regulate activation and solve the problem of insufficient spiking rates activation. Multi-bit spiking reduces the layer by layer transmission of propaga-

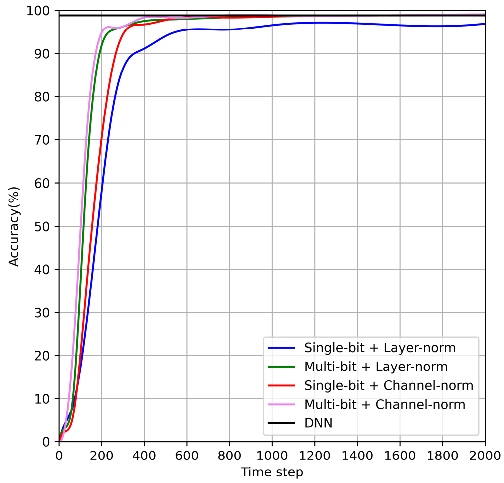

Fig. 3. Performance comparison between our proposed SNN and previous SNN on the UCM dataset.

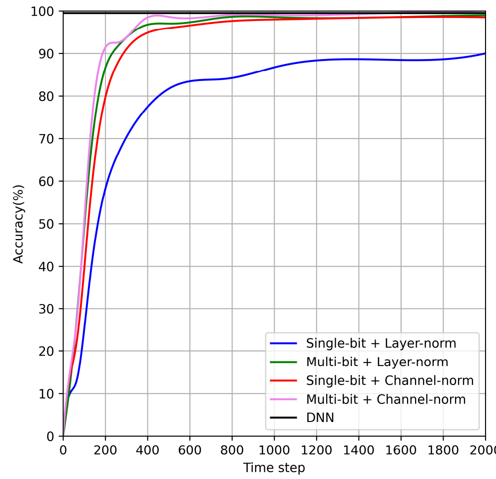

Fig. 4. Performance comparison between our proposed SNN and previous SNN on the WHU-RS dataset.

tion errors by adding a small amount of spiking values.

4. CONCLUSIONS

This paper proposes a spiking neural network suitable for remote sensing imagery scene classification. Based on the previous SNN, a multi-bit-based SNN is designed to reduce layer by layer transmission of propagation errors. Channel normalization is introduced to replace the previous layer normalization, which solves the problem of insufficient spiking rates activation. The experimental results verify the effectiveness of proposed model. In future work, we will explore the application of spiking neural networks in target detection and target segmentation.

5. REFERENCES

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